

CHAPTER 19 · REDOX REACTIONS — COMPLETE REFERENCE SHEET

+n positive -n negative 0 elemental/zero varies variable oxidation state ★ = high priority for test

Element	Ox #	Example Compound	Notes	Element	Ox #	Example Compound	Notes
FIXED RULES — these never change				HALOGENS: Cl, Br, I — variable			
Any pure element	0	H ₂ , O ₂ , Fe, Cu	Elemental state = always 0	Cl in binary salt	-1	NaCl, HCl, CaCl ₂	Default; bonded to metal or H
F in any compound	-1	HF, NaF, CF ₄	Always -1; most electronegative	Cl in ClO ⁻	+1	NaClO	Hypochlorite; bleach
O in most compounds	-2	H ₂ O, CO ₂ , SO ₄ ²⁻	Default for O	Cl in ClO ₂ ⁻	+3	NaClO ₂	Chlorite ion
O in peroxides	-1	H ₂ O ₂ , Na ₂ O ₂	O-O bond present	Cl in ClO ₃ ⁻	+5	KClO ₃	Chlorate ion
O in OF ₂	+2	OF ₂	F outranks O in electronegativity	Cl in ClO ₄ ⁻ ★	+7	KClO ₄	Perchlorate; max Cl state
H with nonmetals	+1	HCl, H ₂ O, HNO ₃	Default for H	Br in binary salt	-1	NaBr, HBr	Default for Br
H with metals	-1	NaH, CaH ₂	Metal hydrides only	I in binary salt	-1	NaI, HI	Default for I
GROUP 1 METALS — always +1				I in IO ₃ ⁻	+5	KIO ₃	Iodate; common in titrations
Li	+1	LiOH, LiCl	Always +1	CARBON (C) — variable			
Na	+1	NaCl, NaOH, Na ₂ O ₂	Always +1	C in CH ₄	-4	CH ₄	Fully reduced; all H bonds
K	+1	KCl, KNO ₃ , KMnO ₄	Always +1	C in CO	+2	CO	Toxic gas
Rb, Cs	+1	RbCl, CsCl	Always +1	C in CO ₂ ★	+4	CO ₂	Most common C oxide
GROUP 2 METALS — always +2				C in CO ₃ ²⁻ ★	+4	CaCO ₃ , Na ₂ CO ₃	Carbonate ion
Mg	+2	MgO, MgCl ₂	Always +2	C in HCOOH	+2	HCOOH	Formic acid
Ca	+2	CaCO ₃ , CaO	Always +2	PHOSPHORUS (P) — variable			
Ba	+2	BaSO ₄	Always +2	P in PH ₃	-3	PH ₃	Phosphine; fully reduced
Be, Sr	+2	BeCl ₂ , SrO	Always +2	P in PO ₄ ³⁻ ★	+5	Na ₃ PO ₄	Phosphate; most important
OTHER FIXED METALS				P in H ₃ PO ₄	+5	H ₃ PO ₄	Phosphoric acid
Al	+3	Al ₂ O ₃ , AlCl ₃	Always +3	TRANSITION METALS — variable (memorize!)			
Zn	+2	ZnO, ZnSO ₄	Always +2	Fe ★	+2	FeO, FeCl ₂ , FeSO ₄	Iron(II) / ferrous
Ag	+1	AgNO ₃ , AgCl	Almost always +1	Fe ★	+3	Fe ₂ O ₃ , FeCl ₃	Iron(III) / ferric; rust
NITROGEN (N) — variable				Cu	+1	Cu ₂ O, CuCl	Cuprous; less common
N in NH ₃	-3	NH ₃	Fully reduced; all H bonds	Cu ★	+2	CuO, CuSO ₄ , Cu(NO ₃) ₂	Cupric; most common
N in N ₂ H ₄	-2	N ₂ H ₄	Hydrazine	Mn	+2	MnO, MnSO ₄	Most stable aqueous form
N in N ₂ O	+1	N ₂ O	Laughing gas	Mn	+4	MnO ₂	Common oxidant
N in NO	+2	NO	Radical; from HNO ₃ reduction	Mn ★	+7	KMnO ₄	Permanganate; strong oxidizer
N in NO ₂ ⁻	+3	NaNO ₂	Nitrite ion	Cr	+3	Cr ₂ O ₃ , CrCl ₃	Most stable Cr state
N in NO ₂	+4	NO ₂	Brown toxic gas	Cr ★	+6	K ₂ CrO ₄ (yellow)	Chromate; basic solution
N in NO ₃ ⁻ ★	+5	KNO ₃ , HNO ₃	Nitrate; max common N state	Cr ★	+6	K ₂ Cr ₂ O ₇ (orange)	Dichromate; acidic solution
SULFUR (S) — variable				Pb	+2	PbO, PbSO ₄	Most common Pb state
S in H ₂ S	-2	H ₂ S	Fully reduced	Pb	+4	PbO ₂	Car battery cathode
S (elemental)	0	S ₈	Pure sulfur	Hg	+1	Hg ₂ Cl ₂	Mercurous; Hg ₂ ²⁺ dimer
S in SO ₃ ²⁻	+4	Na ₂ SO ₃	Sulfite ion	Hg	+2	HgCl ₂	Mercuric; highly toxic
S in SO ₂	+4	SO ₂	Acid rain precursor	Ni	+2	NiO, NiSO ₄	Most common Ni state
S in SO ₄ ²⁻ ★	+6	Na ₂ SO ₄ , H ₂ SO ₄	Sulfate; very common	Co	+2	CoO, CoCl ₂	Pink/red salts
S in SO ₃	+6	SO ₃	Reacts with water → H ₂ SO ₄	Co	+3	Co ₂ O ₃	Common in coordination cmpds
S in S ₂ O ₃ ²⁻	+2	Na ₂ S ₂ O ₃	Thiosulfate; redox titrations	KEY POLYATOMIC ION ELEMENTS			
				N in NH ₄ ⁺ ★	-3	NH ₄ Cl, (NH ₄) ₂ SO ₄	Ion overall charge = +1
				O in OH ⁻ ★	-2	NaOH, Ca(OH) ₂	Hydroxide; ion charge = -1
				O in O ₂ ²⁻	-1	H ₂ O ₂ , Na ₂ O ₂	Peroxide; each O = -1
				C in CN ⁻	+2	NaCN, KCN	Cyanide; N = -3

CHAPTER 19 · REDOX CONCEPTS, BALANCING & ELECTROCHEMISTRY

Species / Concept	Formula / Example	Ox #	Notes / Details
CORE DEFINITIONS			
Redox reaction			Transfer of electrons between substances; ox numbers change
Oxidation			Atom LOSES electrons → ox number becomes MORE positive (OIL)
Reduction			Atom GAINS electrons → ox number becomes LESS positive (RIG)
Reducing agent			Causes reduction of another — itself gets OXIDIZED
Oxidizing agent			Causes oxidation of another — itself gets REDUCED
ELECTROCHEMICAL CELLS			
Electrolytic cell			Uses ELECTRICITY to FORCE nonspontaneous redox reaction
Voltaic cell			SPONTANEOUS redox reaction PRODUCES electricity (batteries)
Cathode			Reduction occurs; (+) in voltaic, (–) in electrolytic
Anode			Oxidation occurs; (–) in voltaic, (+) in electrolytic
Salt bridge			Allows ion flow between half-cells; prevents mixing
Electrolysis			Forcing nonspontaneous redox with electrical current
Electroplating			Object to plate = cathode; plating metal = anode
BALANCING REDOX: 5-STEP METHOD			
Step 1			Assign oxidation numbers to all atoms
Step 2			Identify which atoms are oxidized (increase) and reduced (decrease)
Step 3			Draw lines between ox/red atoms; note net change in ox number
Step 4			Multiply pairs by coefficients so total Δox = total Δred
Step 5			Use inspection to finish balancing atoms and charge
Species / Concept	Formula / Example	Ox #	Notes / Details
PPT EXAMPLES TO KNOW			
Ex 19-1a (NOT redox)	$\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$		No ox numbers change — NOT a redox reaction
Ex 19-1b (redox)	$2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$		Cl: +5 → –1; O: –2 → 0 — YES redox
Ex 19-2	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$		Na oxidized (0 → +1); H reduced (+1 → 0)
Ex 19-4 balanced	$3\text{HCl} + 2\text{HNO}_3 \rightarrow 3\text{HOCl} + 2\text{NO} + \text{H}_2\text{O}$		Cl: –1 → +1 (ox); N: +5 → +2 (red)
Ex 19-5 half-rxns	$3\text{Ca} \rightarrow 3\text{Ca}^{2+} + 6\text{e}^-$		Anode (oxidation)
	$2\text{Fe}^{3+} + 6\text{e}^- \rightarrow 2\text{Fe}$		Cathode (reduction)
Rusting	$4\text{Fe}(\text{OH})_2 + \text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 4\text{H}_2\text{O}$		Fe: +2 → +3 oxidized; O ₂ reduced
Voltaic (Zn/Cu)	$\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$		~1.1 V; Zn = anode; Cu = cathode
QUICK RULES SUMMARY			
★ = high-priority	Know these cold		Fe ₂ O ₃ , CuSO ₄ , KMnO ₄ , NO ₃ [–] , CO ₃ ^{2–} , SO ₄ ^{2–} , ClO ₄ [–] , PO ₄ ^{3–} , OH [–] , NH ₄ ⁺
Ox # of ion			Sum of all ox numbers in polyatomic ion = ion's overall charge
Neutral compound			Sum of all ox numbers = 0
Redox check			If ANY ox number changes between reactants and products → it IS redox
OIL RIG mnemonic			Oxidation Is Loss of electrons — Reduction Is Gain of electrons

OIL RIG: Oxidation Is Loss · Reduction Is Gain | Cathode = Reduction = C·R | Anode = Oxidation = A·O | Oxidation numbers in neutral compound always sum to 0